

WALMART

Business Intelligence & Analytics

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Dataset Overview | <https://www.kaggle.com/datasets/najir0123/walmart-10k-sales-datasets>

Column	Description	Example Value
invoice_id	Unique transaction identifier	WALM003
Branch	Store branch ID	WALM048
City	City of the branch	San Antonio
category	Product category	Health and beauty
unit_price	Price per unit	\$74.69
quantity	Items purchased	7
date	Transaction date	05-01-19
time	Transaction time	13:08:00
payment_method	Method of payment	Ewallet, Cash, Credit card
rating	Customer satisfaction score	9.1
profit_margin	Profit margin ratio	0.48

Business Questions (9 Questions)

The following questions were defined as the initial set of analytical objectives for the Walmart dataset.

Q1

Question: What are the different payment methods, and how many transactions and items were sold with each method?

Purpose: This helps understand customer preferences for payment methods, aiding in payment optimization strategies.

Q2

Question: Which category received the highest average rating in each branch?

Purpose: This allows Walmart to recognize and promote popular categories in specific branches, enhancing customer satisfaction and branch-specific marketing.

Q3

Question: What is the busiest day of the week for each branch based on transaction volume?

Purpose: This insight helps in optimizing staffing and inventory management to accommodate peak days.

Q4

Question: What is the average, minimum, and maximum ratings for each category in each city?

Purpose: This data can guide city-level promotions, allowing Walmart to address regional preferences and improve customer experiences.

Q5

Question: What is the total profit for each category, ranked from highest to lowest?

Purpose: Identifying high-profit categories helps focus efforts on expanding these products or managing pricing strategies effectively.

Q6

Question: What is the most frequently used payment method in each branch?

Purpose: This information aids in understanding branch-specific payment preferences, potentially allowing branches to streamline their payment processing systems.

Q7

Question: How many transactions occur in each shift (Morning, Afternoon, Evening) across branches?

Purpose: This insight helps in managing staff shifts and stock replenishment schedules, especially during high-sales periods.

Q8

Question: What is the average unit price for each product category, and how does it correlate with customer ratings?

Purpose: Understanding the price-rating relationship helps Walmart determine whether premium-priced categories deliver perceived value, guiding pricing and quality improvement strategies.

Q9

Question: How does customer rating vary by time of day (Morning, Afternoon, Evening) across all branches?

Purpose: Identifying the time periods with the highest satisfaction scores helps Walmart schedule premium service resources and promotions when customers are most receptive.